

# Power BI Assignment 8

## ✿ 1. Power Pivot Components and Their Tasks

**Power Pivot** is an Excel add-in (and part of Power BI) used for data modeling and advanced analytics. It allows users to create relationships, perform complex calculations, and analyze large datasets efficiently.

### ◆ Main Components and Their Tasks:

| Component                           | Description                                                                       | Task / Function                                                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Data Model                          | The core of Power Pivot where all tables, relationships, and calculations reside. | Store and manage imported data from multiple sources.                                                      |
| Data View                           | Displays table data like an Excel sheet.                                          | View, inspect, and clean data.                                                                             |
| Diagram View                        | Shows the relationships between tables visually.                                  | Create and manage table relationships (1:many, many:1).                                                    |
| Calculation Area (DAX)              | Section where you create formulas.                                                | Build <b>measures, calculated columns</b> , and <b>KPIs</b> using <b>DAX (Data Analysis Expressions)</b> . |
| Relationships                       | Define how data in different tables connect.                                      | Enable cross-table analysis (e.g., linking Sales to Product).                                              |
| PivotTable / PivotChart Integration | Integration with Excel visual tools.                                              | Analyze and visualize data interactively.                                                                  |

## 🖥️ 2. Different Views in Power BI Desktop

Power BI Desktop provides **three main views**, each serving a unique purpose in the report development process.

| View Name                 | Icon                                                                                | Purpose / Function                                       | Working                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Report View               |  | Used to design dashboards and visual reports.            | You can add visuals (charts, KPIs, maps), apply filters, and interact with slicers. It's the main presentation view.        |
| Data View                 |  | Displays underlying data tables imported into the model. | You can see raw data, create calculated columns, and use DAX for computations.                                              |
| Model (Relationship) View |  | Shows how tables are connected via relationships.        | Manage relationships between fact and dimension tables, define cardinality (1:* or *:1), and organize data schema visually. |

### ◆ Summary Visualization

[ Report View ] → Build Visuals

[ Data View ] → Explore / Calculate Data

[ Model View ] → Define Relationships

---

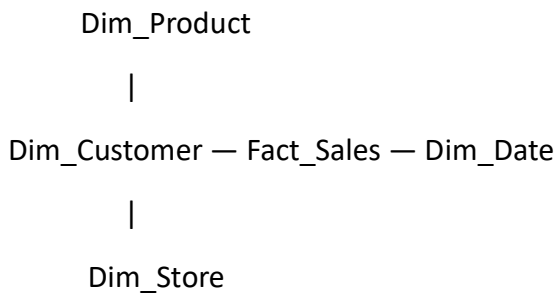
### 3. Star and Snowflake Schema

#### Star Schema

##### Definition:

A **Star Schema** consists of a **central fact table** linked directly to **dimension tables**. It is simple, denormalized, and optimized for fast querying.

##### Structure:



##### Example Explanation:

- **Fact Table:** Sales (contains SalesAmount, Quantity, etc.)
- **Dimension Tables:** Product, Customer, Store, Date (describe each aspect of sales)

##### Advantages:

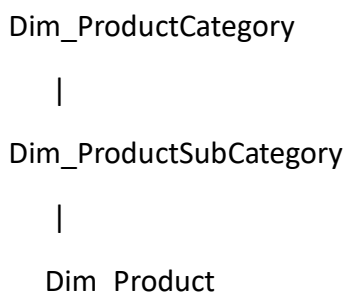
- Faster performance
  - Easy to understand and maintain
  - Ideal for Power BI and OLAP models
- 

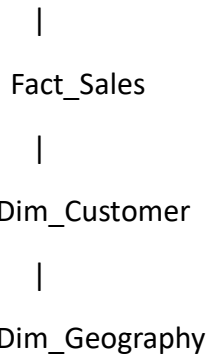
#### Snowflake Schema

##### Definition:

A **Snowflake Schema** is a more normalized version of a Star Schema where **dimension tables are split into sub-dimensions**.

##### Structure:





**Example Explanation:**

- The Product dimension is broken into **Product → Subcategory → Category**.
- This reduces data redundancy but increases query complexity.

**Advantages:**

- Saves storage space
- Maintains data integrity (less duplication)

**Disadvantages:**

- More complex joins
- Slightly slower performance

 **4. Dimension Table vs Fact Table**

| Aspect      | Dimension Table                                                               | Fact Table                                            |
|-------------|-------------------------------------------------------------------------------|-------------------------------------------------------|
| Purpose     | Stores descriptive or categorical information (the "who, what, where, when"). | Stores quantitative or numeric data (the "how much"). |
| Contains    | Textual fields like ProductName, CustomerName, Region, Date.                  | Numeric measures like SalesAmount, Quantity, Profit.  |
| Primary Key | Unique identifier (e.g., ProductID).                                          | Composite key made of foreign keys from dimensions.   |
| Size        | Usually small.                                                                | Usually large.                                        |
| Example     | Product, Customer, Store, Date.                                               | Sales, Revenue, Transactions.                         |

**Illustration:**

